

PH101: 2023

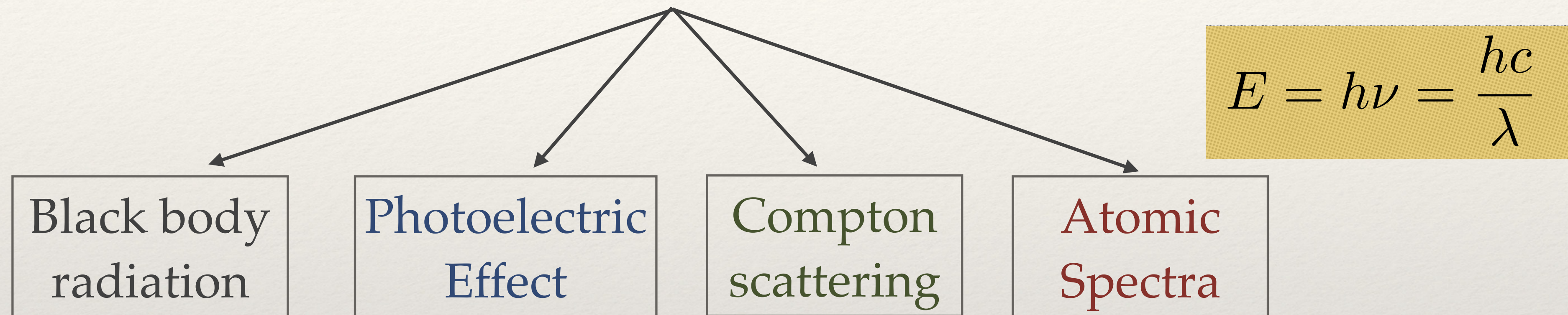
Quantum Mechanics

Lecture 2

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Recap

- We talked about a series of experiments which could only be explained by particle nature of wave.



- De Broglie's hypothesis on wave particle duality: each matter particle also shows wave nature. The relation between wavelength and momentum was postulated via Planck's constant.

$$\lambda = \frac{h}{p}$$

But we need to experimentally verify matter wave!

Interference: Superposition of waves

Consider the amplitude of a sinusoidal wave travelling along x-axis:

$$W_1(x, t) = A \cos(kx - \omega t)$$
$$k = \frac{2\pi}{\lambda} \quad \text{Wavenumber}$$
$$\omega = 2\pi f \quad \text{Angular frequency}$$

Consider a second wave of same amplitude and frequency, but different phase.

$$W_2(x, t) = A \cos(kx - \omega t + \varphi)$$

Superposition of the waves:

$$W_1 + W_2 = A[\cos(kx - \omega t) + \cos(kx - \omega t + \varphi)].$$

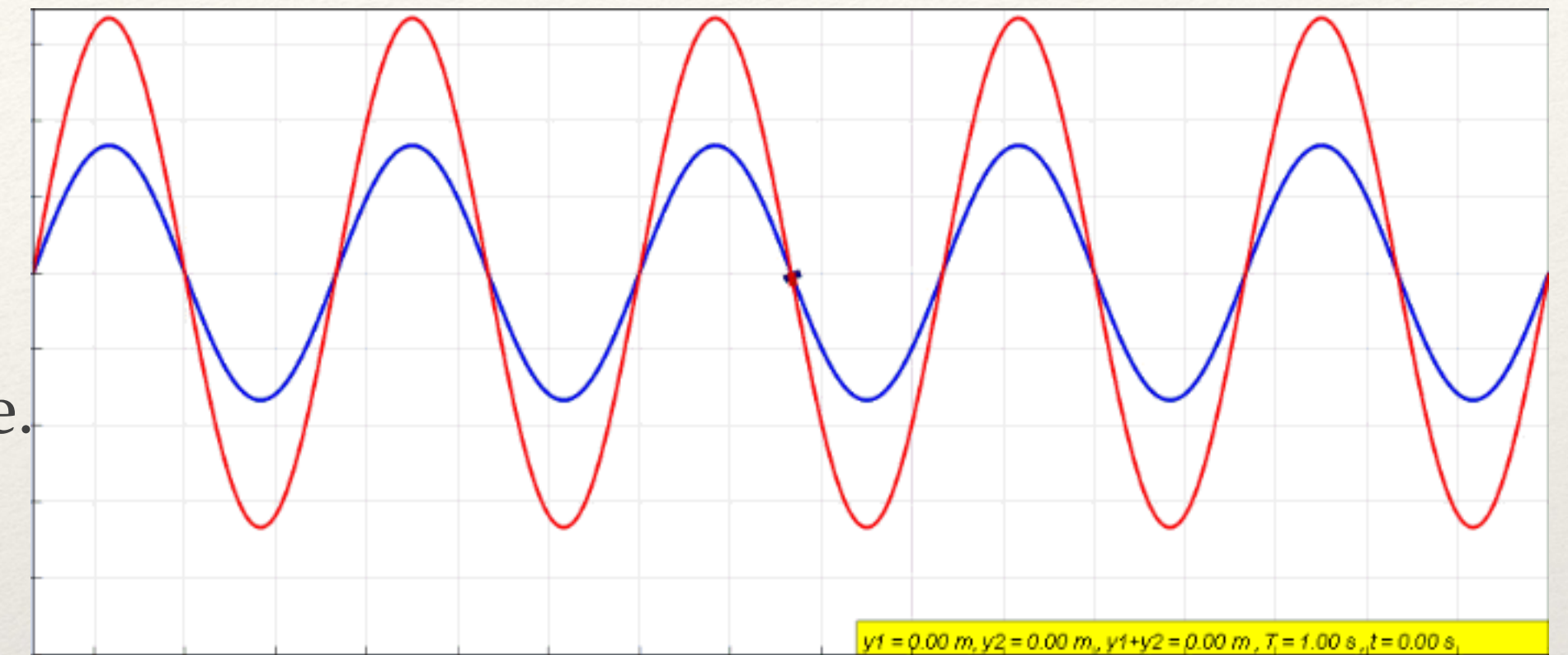
$$W_1 + W_2 = 2A \cos\left(\frac{\varphi}{2}\right) \cos\left(kx - \omega t + \frac{\varphi}{2}\right).$$

$$\cos a + \cos b = 2 \cos\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right)$$

Constructive Interference

$$W_1 + W_2 = 2A \cos(kx - \omega t)$$

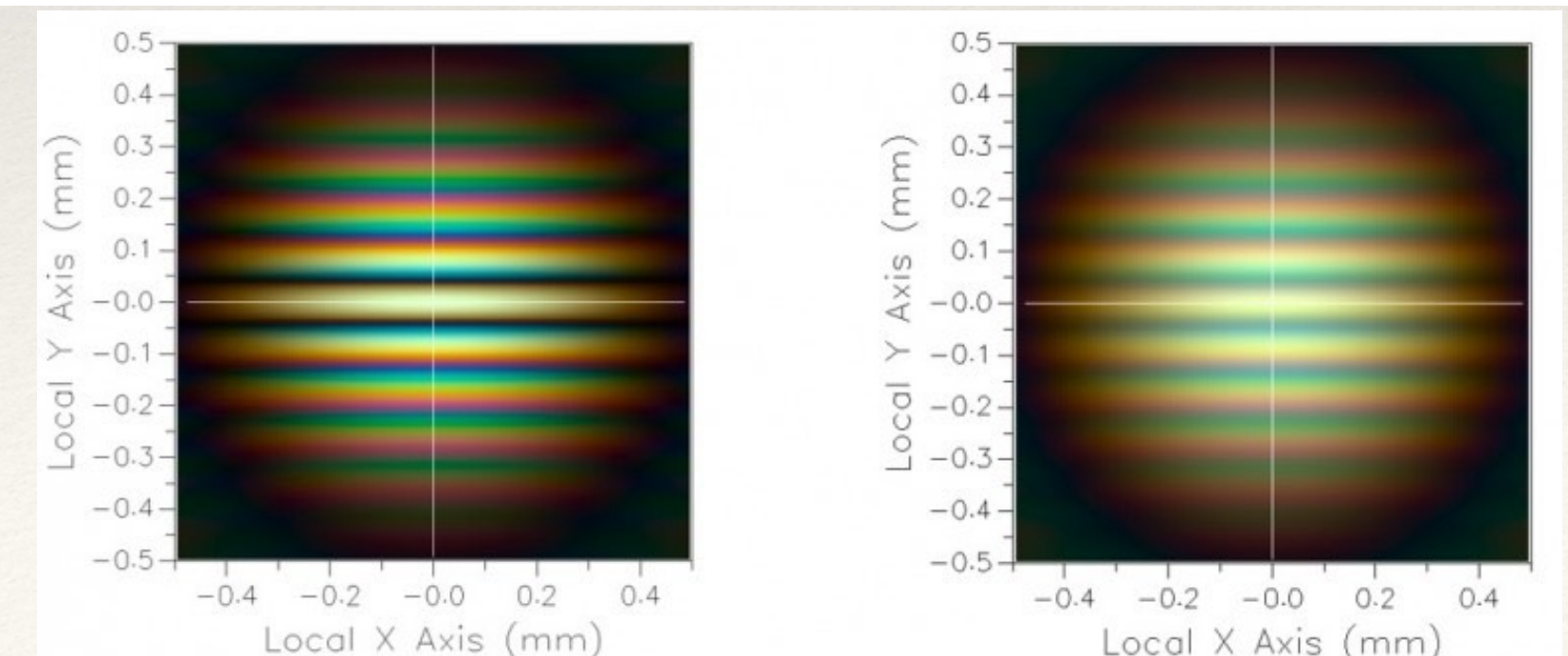
even multiple of π : $\varphi = \dots, -4\pi, -2\pi, 0, 2\pi, 4\pi, \dots$ then $|\cos(\varphi/2)| = 1$,



Destructive Interference

$$W_1 + W_2 = 0$$

odd multiple of π : $\varphi = \dots, -3\pi, -\pi, \pi, 3\pi, 5\pi, \dots$ then $\cos(\varphi/2) = 0$



Double slit Experiments with Bullets

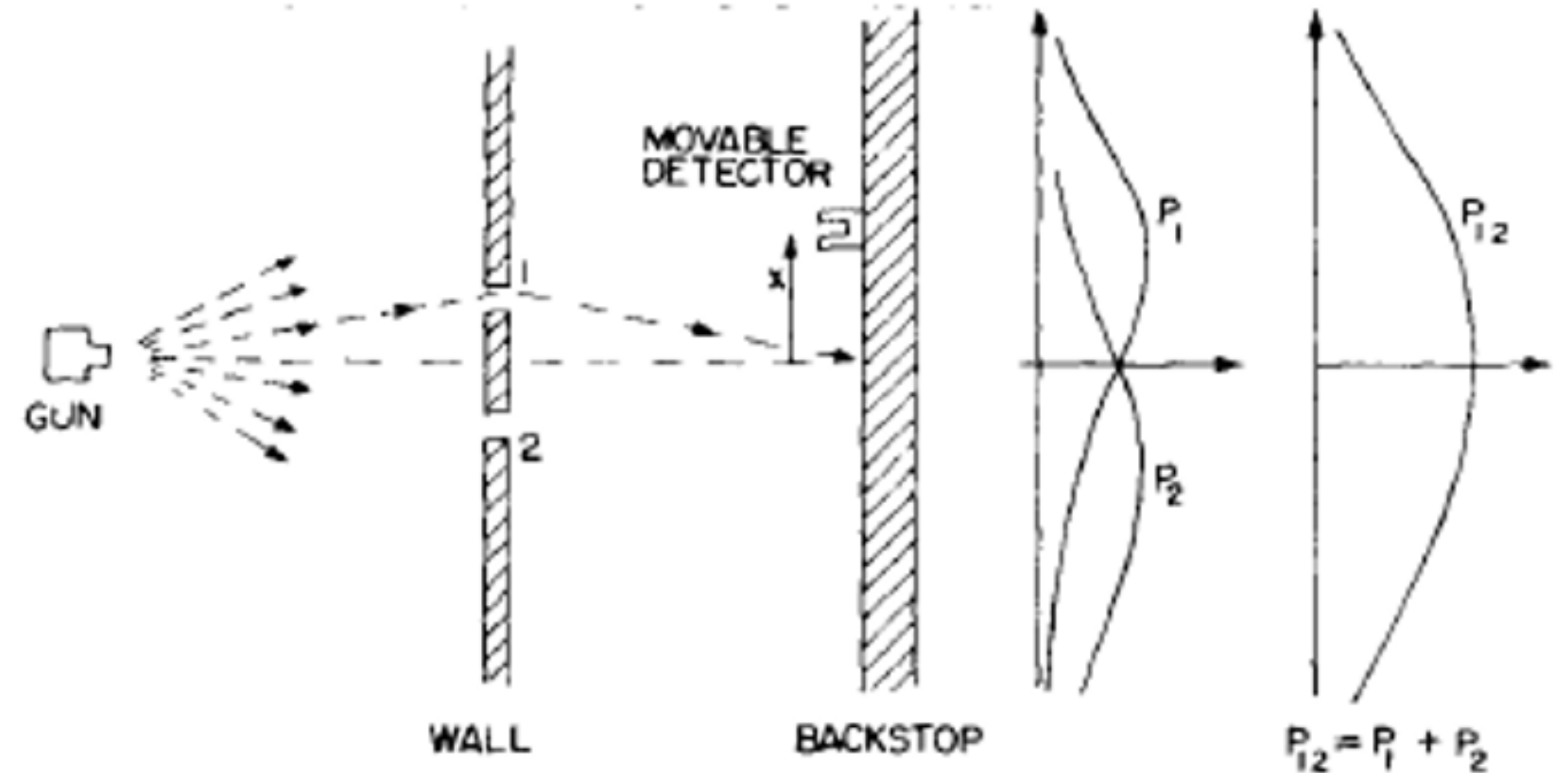
Experiment with a gun (bullets as particles):

Detector counts number of bullets.

P_1 : probability when hole 2 is closed.

P_2 : probability when hole 1 is closed.

P_{12} : probability when both holes are open.



$$P_{12} = P_1 + P_2$$

Result: No interference pattern

Young did this double slit experiment around 1801 to show the wave properties of light. So it is sometimes called Young's double slit experiment.

Experiment with wave

Experiment with water wave:

Detector records intensity of the wave.

I_1 : probability when hole 2 is closed.

I_2 : probability when hole 1 is closed.

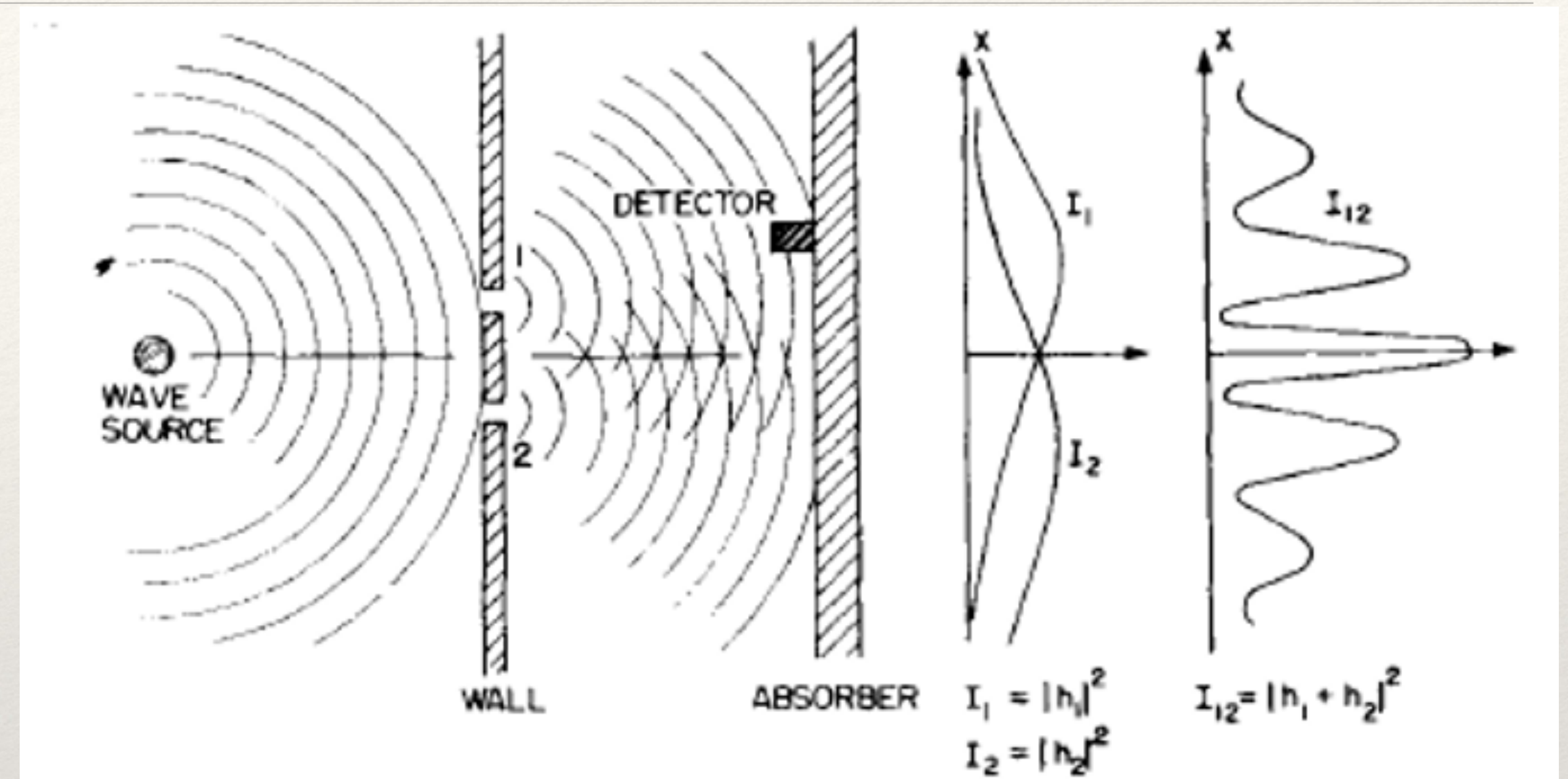
I_{12} : probability when both holes are open.

$$I_{12} \neq I_1 + I_2$$

$$|h_1 + h_2|^2 = |h_1|^2 + |h_2|^2 + 2|h_1||h_2|\cos\delta,$$

$$I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\delta.$$

Shows up interference pattern

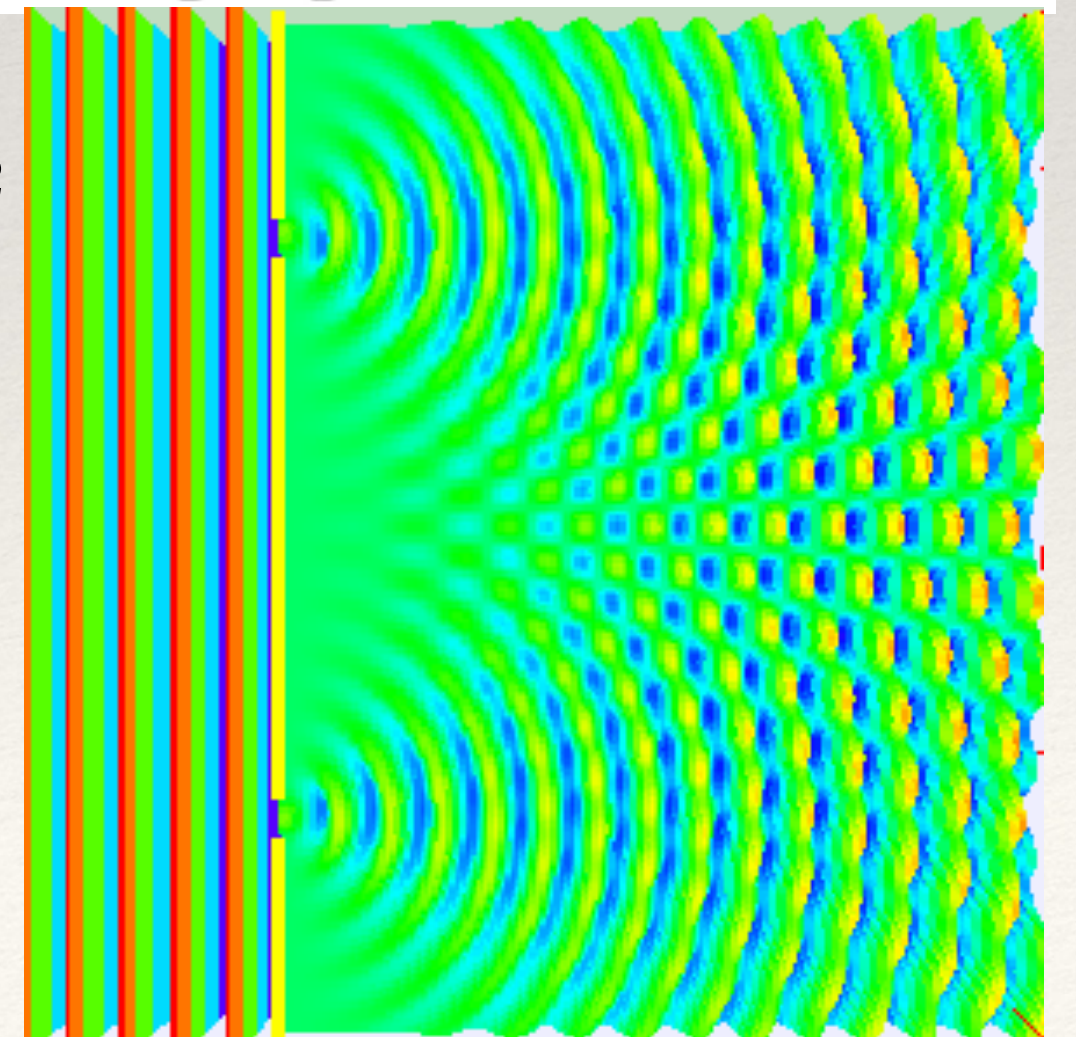


$$\delta = \{0, 2\pi, 4\pi \dots\} : I_{12} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Constructive interference

$$\delta = \{0, \pi, 3\pi \dots\} : I_{12} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Destructive interference



Experiment with Electron

Detector counts number of electrons.

P_1 : probability when hole 2 is closed.

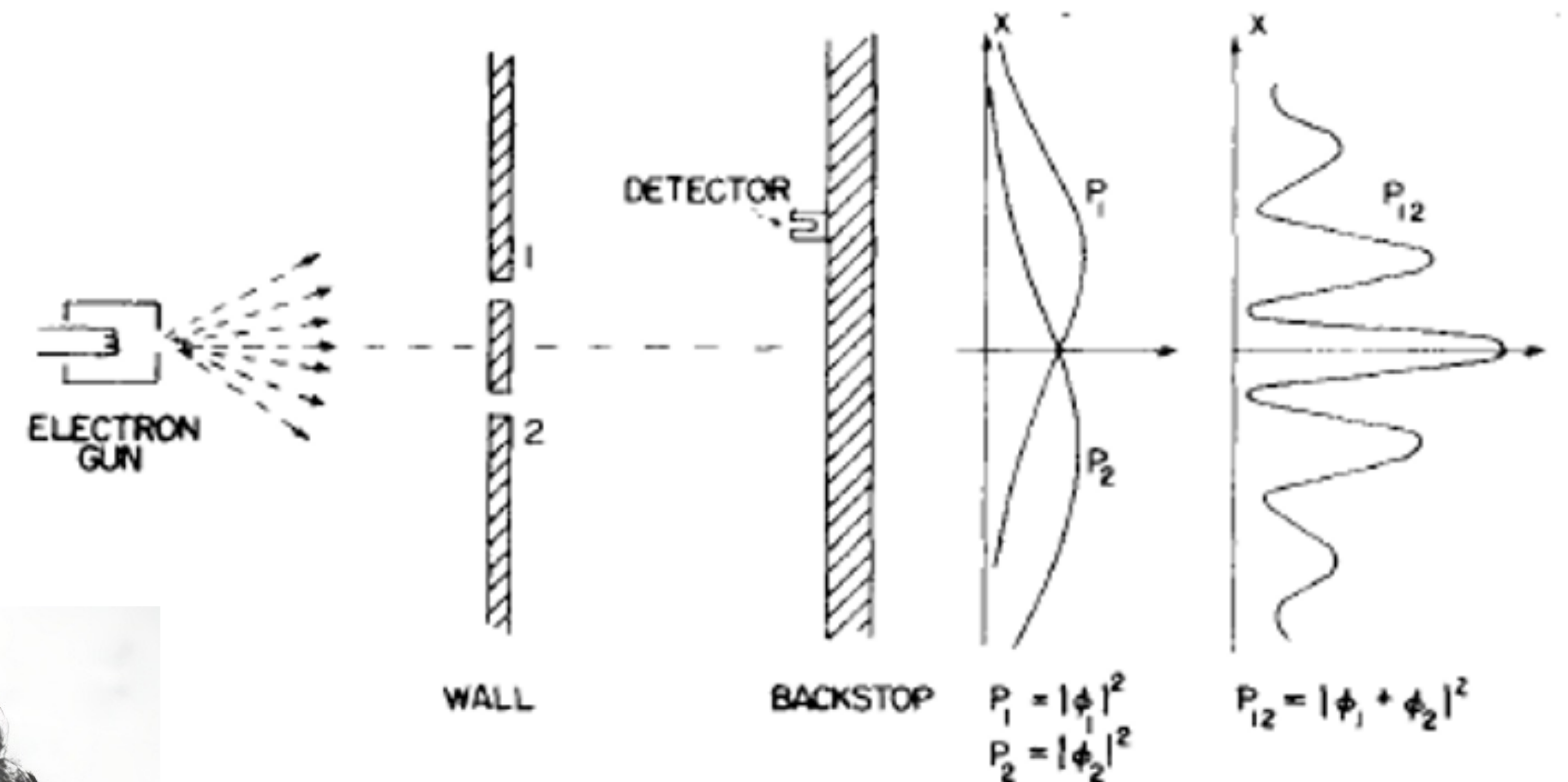
P_2 : probability when hole 1 is closed.

P_{12} : probability when both holes are open.

$$P_{12} \neq P_1 + P_2$$

Results an interference

While the electrons arrive in lumps, like particles, and probability of arrival of these lumps is distributed like the intensity of a wave. In this sense, the electrons behave like wave:
A proposal by Richard Feynman



Although some earlier attempts were made (1964), this could be done experimentally only in 2012 by Stefano Frabboni and collaborators!

Verification of matter wave

Electron Diffraction

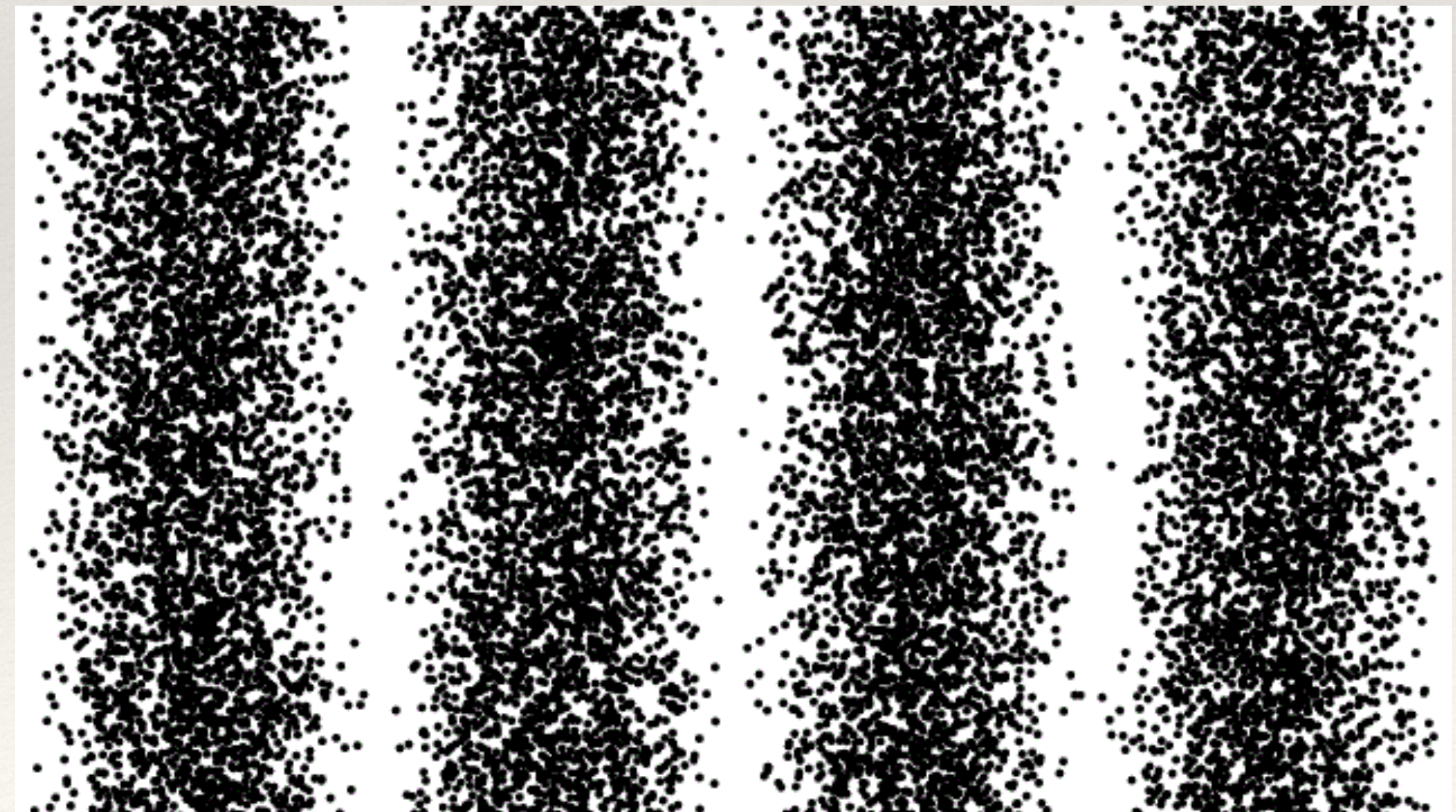
Electron diffraction refers to changes in the direction of **electron beams** due to **atomic** structures generating **diffraction patterns** which are widely used for analysis of materials. Electron diffraction can also refer to derived experimental techniques used for **material characterization**. This technique is similar to **X-ray** and **neutron diffraction**.~ Depends on the wave properties of matter.

Davisson and Germer during 1923-1927 in Bell labs did show the diffraction pattern of the electron from Nickel plate verifying the matter wave.

This in turn verified the De Broglie's hypothesis.

At this site, the original location of Bell Telephone Laboratories, C. J. Davisson and L. H. Germer in 1927 performed the first direct demonstration of the wave-like behavior of elementary particles, predicted by L. de Broglie in 1923. The Davisson-Germer experiment provided crucial empirical evidence for the validity of the then rapidly evolving theory of quantum mechanics. In those years and subsequently many important scientific and technological discoveries were made at the same laboratory.

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De Broglie's hypothesis for massive particle

The argument goes like this:

For a massive particle, in its rest frame, $E = m_0 c^2 = h\nu_0$

While from the laboratory frame, $E = mc^2 = h\nu$

Thus we recreate the hypothesis for massive particles!

Following, $m = \frac{m_0}{(1 - \beta^2)^{1/2}} \rightarrow \nu = \frac{\nu_0}{(1 - \beta^2)^{1/2}}$

Some characteristic vibration in the rest frame: $\xi_0 \sim \sin 2\pi\nu_0 t_0$

Substituting

$$t_0 = \frac{1}{(1 - \beta^2)^{1/2}} \left(t - \frac{vx}{c^2} \right)$$

To evaluate the same in lab frame, we need to use Lorentz transformation:

Therefore, $\xi(x, t) \sim \sin \frac{2\pi\nu_0}{(1 - \beta^2)^{1/2}} \left(t - \frac{vx}{c^2} \right) \rightarrow \xi(x, t) \sim \sin 2\pi\nu \left(t - \frac{x}{w} \right)$

$$\omega = \frac{c^2}{v}$$

Velocity of the phase

The wavelength $\lambda = \frac{\omega}{\nu} = \frac{c^2}{v} \frac{(1 - \beta^2)^{1/2}}{\nu_0} = \frac{hc^2}{v} \frac{(1 - \beta^2)^{1/2}}{h\nu_0} = \frac{hc^2}{v} \frac{(1 - \beta^2)^{1/2}}{m_0 c^2} = \frac{h}{mv} = \frac{h}{p}$

Uncertainty principle



Werner Heisenberg
Nobel prize: 1932

$$\Delta p \Delta x \geq \frac{h}{4\pi}$$

Planck's constant h

Uncertainty in momentum Δp

Uncertainty in position Δx

This provides a departure from the deterministic approach of classical physics

1927: Warner Heisenberg proposed, that position and momentum can't be measured simultaneously in a precise way!

The essence of uncertainty is that the act of measurement influences the particle, gives it a kick and it no longer remains where it was.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

$$\hbar = \frac{h}{2\pi}$$

Values		
Constant	SI units	Units with eV
h	$6.626\,070\,15 \times 10^{-34} \text{ J} \cdot \text{Hz}^{-1}$ ^[2]	$4.135\,667\,696... \times 10^{-15} \text{ eV} \cdot \text{Hz}^{-1}$ ^[5]
$\hbar$	$1.054\,571\,817... \times 10^{-34} \text{ J} \cdot \text{s}$ ^[6]	$6.582\,119\,569... \times 10^{-16} \text{ eV} \cdot \text{s}$ ^[7]
hc	$1.986\,445\,86... \times 10^{-25} \text{ J} \cdot \text{m}$	$1.239\,841\,98... \text{ eV} \cdot \mu\text{m}$
$\hbar c$	$3.161\,526\,77... \times 10^{-26} \text{ J} \cdot \text{m}$	$0.197\,326\,9804... \text{ eV} \cdot \mu\text{m}$

Heisenberg's thought Experiment

To see an electron, we illuminate it by throwing light on it, which scatters to microscope viewer

Resolution of two points at x and $x+\Delta x$ require angular separation:

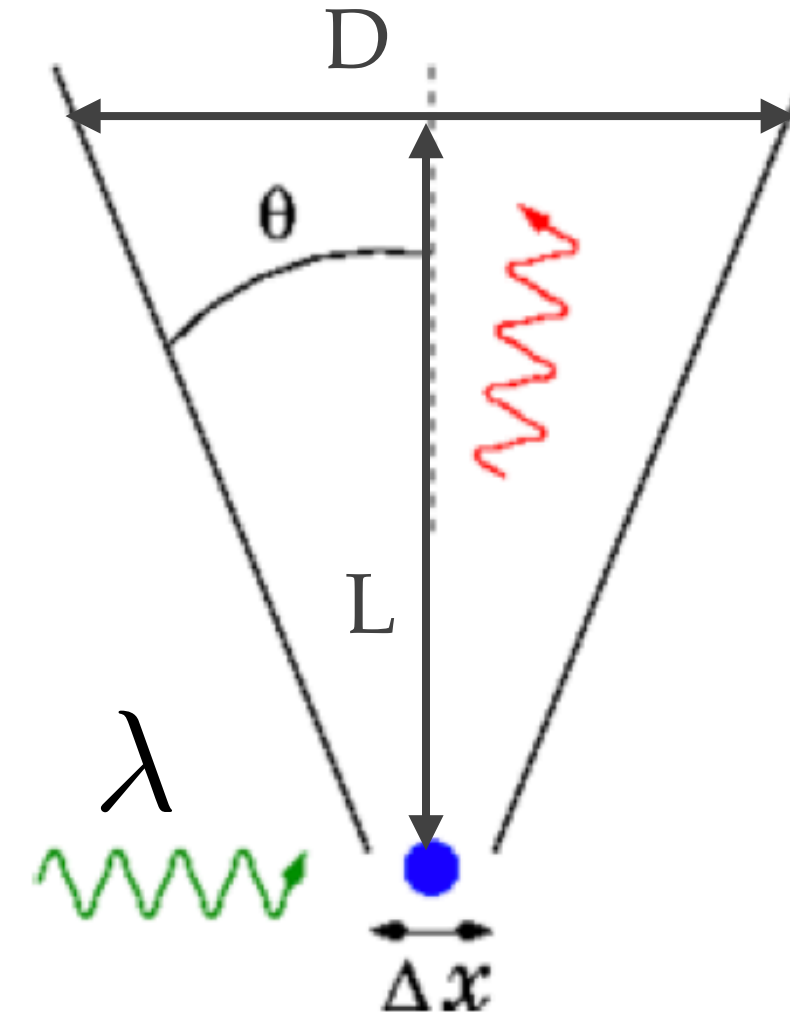
$$\Delta\phi = \frac{\Delta x}{L}$$

The smallest angular separation that can be measured (Diffraction pattern) by a lens of diameter D is:

$$\Delta\phi_{min} \approx \frac{\lambda}{D}$$

$$\frac{\Delta x}{L} \sim \frac{\lambda}{D} \rightarrow$$

$$\Delta x \approx \frac{\lambda}{D/L} \approx \frac{\lambda}{\sin \theta}$$



Smaller the wave-length, the finer is the position measurement.

The uncertainty in photon momentum along x-axis is from $-p \sin \theta$ to $p \sin \theta$: $2p \sin \theta$

Then electron's momenta along x-axis is uncertain by the same amount:

$$\Delta p_x = 2p \sin \theta = 2 \frac{h}{\lambda} \sin \theta$$

The larger the wavelength, the lesser is the uncertainty in the momentum

$$\Delta x \Delta p_x = 2h$$

Example: Stability of Atom

Position-momentum uncertainty can be used to find the bound state energy of Hydrogen atom

Total energy of electron in the field of proton:

$$E = \underbrace{\left(\frac{p^2}{2m}\right)}_{\text{Kinetic energy}} - \underbrace{\left(\frac{e^2}{(4\pi\epsilon_0)r}\right)}_{\text{Electrostatic potential energy}}$$

Classically, there is no lower limit to E as r can be as small as possible. No bound state !

However, quantum mechanically,

p can be measured with uncertainty $\Delta p \sim p$
 r can be measured with uncertainty $\Delta r \sim r$

$$\Delta r \Delta p \sim \hbar \rightarrow rp \sim \hbar$$

Substituting,

$$E \simeq \frac{\hbar^2}{2mr^2} - \frac{e^2}{(4\pi\epsilon_0)r}$$

Minimum energy can be found out by,

$$\frac{\partial E}{\partial r} = 0$$

$$r_0 \simeq \frac{(4\pi\epsilon_0)\hbar^2}{me^2} = a_0$$

Bohr radius

Substituting back in E, we find,

$$E_0 \simeq -\frac{e^2}{(4\pi\epsilon_0)2a_0}$$

Ground state energy of Hydrogen.

This is the minimum energy required to separate an electron out of the bound state.

Energy-time uncertainty

Consider an electron moving along x-axis.

For a non-relativistic free particle:

$$E = \frac{p_x^2}{2m}$$

The uncertainty in momentum measurement gives rise to uncertainty in energy:

$$\Delta E = (p_x/m)\Delta p_x = v_x \Delta p_x.$$

Recoil velocity of the electron after scattering with photon

If the time interval required for the measurement is Δt



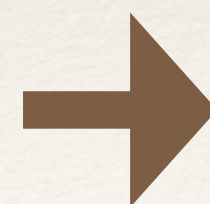
$$\Delta x = v_x \Delta t.$$

$$v_x = \frac{\Delta x}{\Delta t}$$

$$\Delta E \Delta t = \Delta p_x \Delta x.$$

Following,

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$



$$\Delta E \Delta t \geq \hbar/2$$

A similar uncertainty exists in terms of energy and time

Newton's law and uncertainty

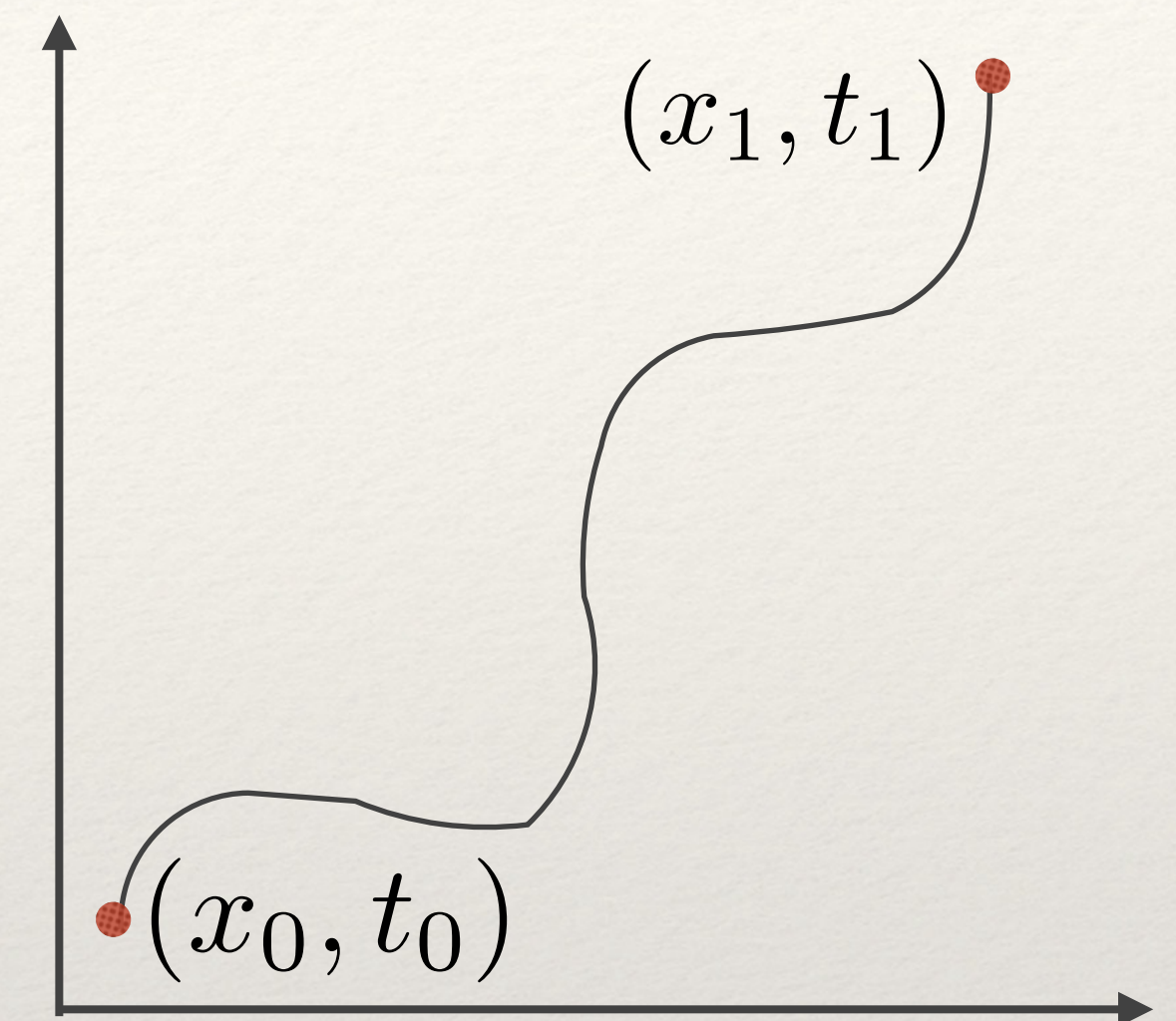
$$F = m \frac{d^2 x}{dt^2} \quad \rightarrow \quad \int_{t=0}^t \frac{F}{m} dt = \int \frac{d}{dt} \left(\frac{dx}{dt} \right) dt$$

$$\frac{dx}{dt} = \frac{dx}{dt} \Big|_{t=0} + \int_{t=0}^t \frac{F}{m} dt = u + at$$

$$\boxed{\frac{dx}{dt} \Big|_{t=0} = u, \quad \frac{F}{m} = a}$$

$$\rightarrow \int_{t=0}^t \frac{dx}{dt} dt = ut + \frac{1}{2} at^2$$

$$x(t) = \boxed{x|_{t=0}} + ut + \frac{1}{2} at^2$$



Solving the dynamics of a classical particle requires the knowledge of position and momentum at some initial time $t=0$.

However, due to uncertainty in quantum world,

Absent the information of $x|_{t=0}$ and $\frac{dx}{dt} \Big|_{t=0}$, following uncertainty relation, we can't infer the dynamics anymore via this relation.

Classical Limit

Consider the phase space, coordinate-momentum space!

Quantum mechanically, the minimum volume that a particle occupy has size:

$$\Delta x \Delta p_x \approx h$$

However, classically, it should be a point and any position and any momentum are equally probable.

Therefore, we can say, classical limit is obtained when $h \rightarrow 0$

Now, bring back De Broglie's relation of wave particle duality:

$$\lambda = \frac{h}{p} \rightarrow \lambda p = h$$

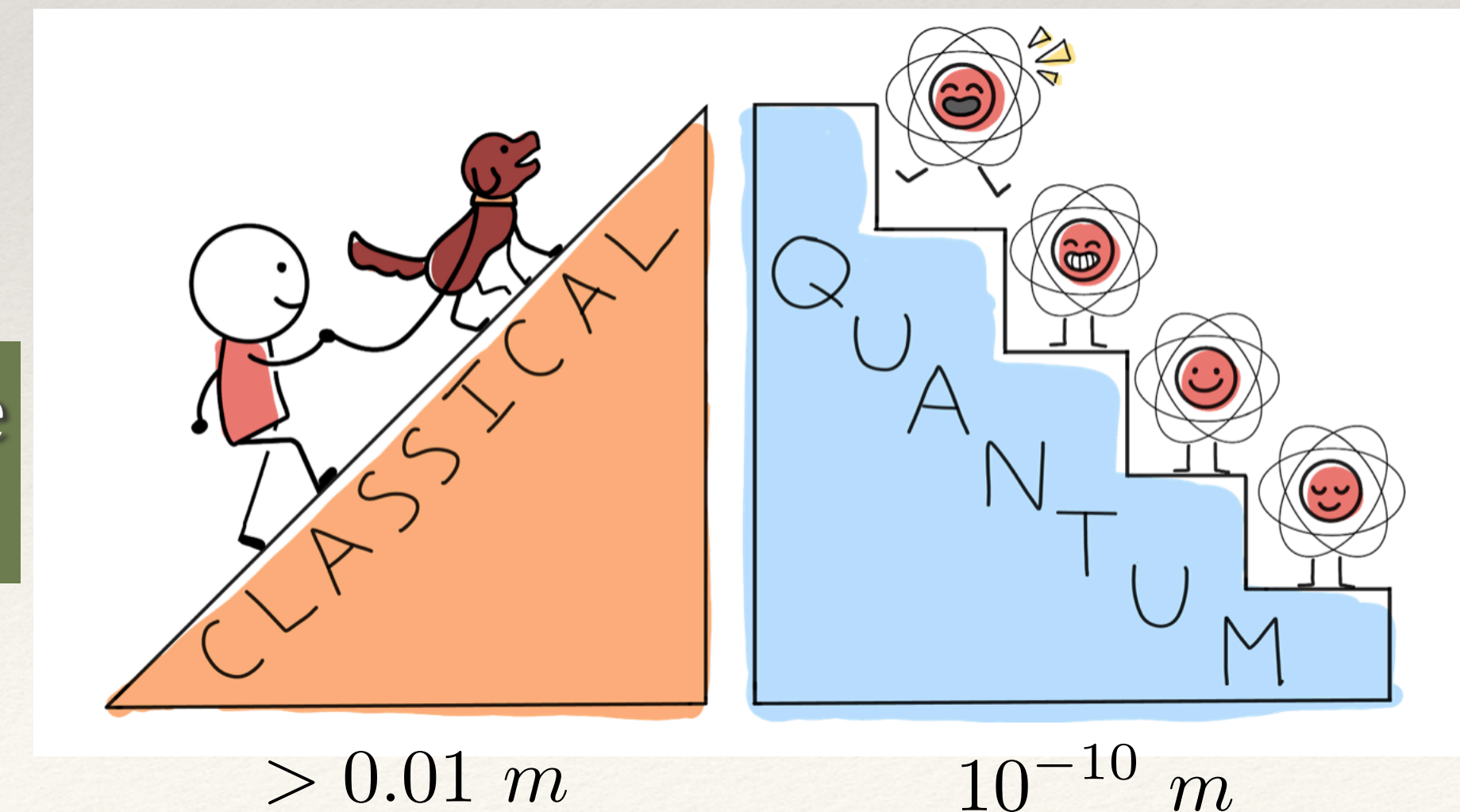
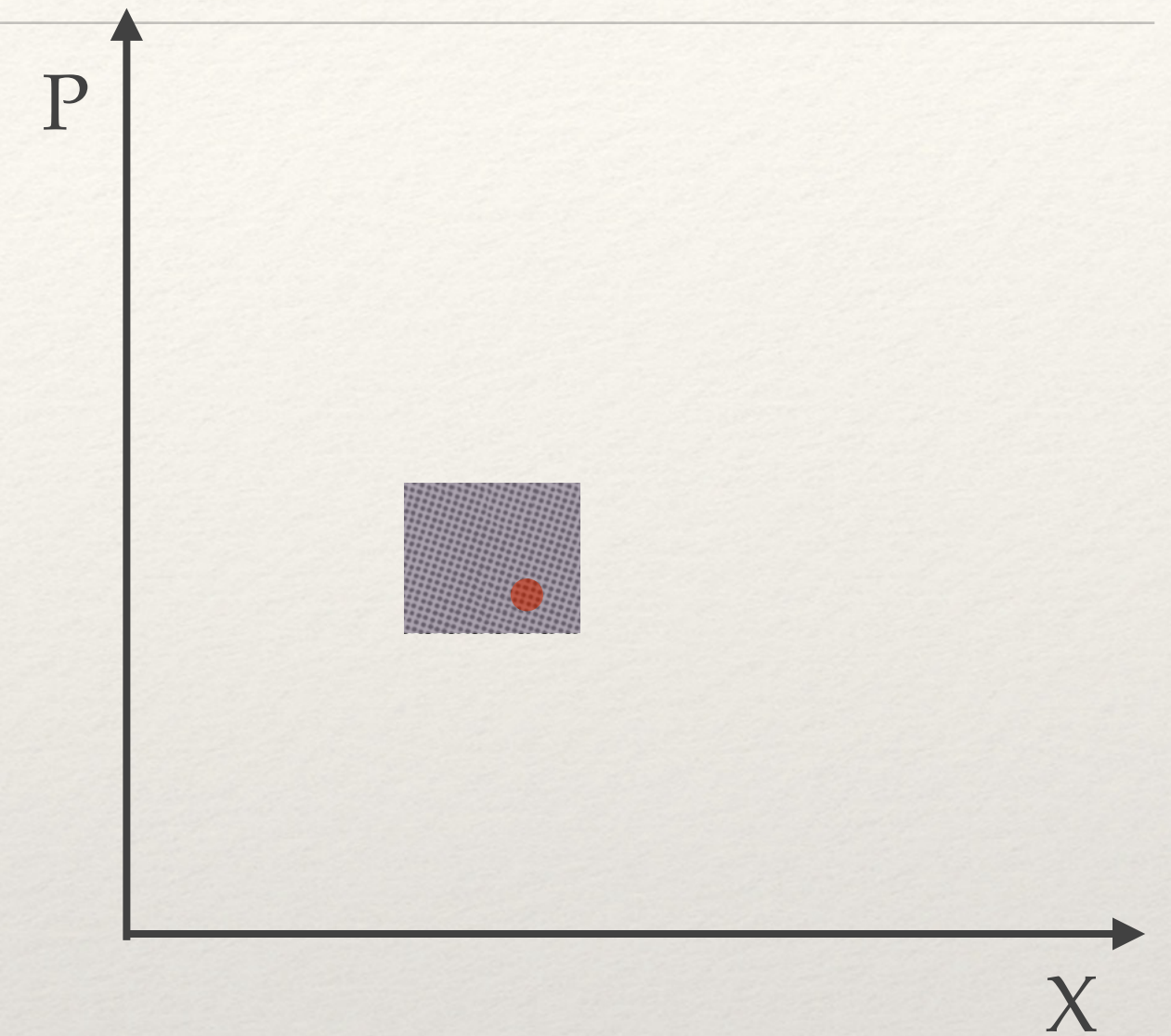
Therefore in the classical limit, $\lambda p \rightarrow 0 \rightarrow \lambda \rightarrow 0$ Or $p \rightarrow 0$

Therefore, Planck's constant decide upon where we should have quantum regime.

Only particle nature!

Only wave nature!

As we normally see



Summary

- Uncertainty principle, or the lack of precise simultaneous measurement of position and momentum marks a significant departure from the deterministic approach of classical mechanics.

We need to find a probabilistic interpretation for the quantum physics